

Environmental Science

Vedic Perspective

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Science and Technology in Ancient Indian Texts



Introduction: The Vedas & Environment

Key Facts

- Vedas are the first texts in the library of mankind
- Most precious Indian heritage — universally acknowledged
- Cover knowledge both physical and spiritual (Manusmṛiti)
- Vedic views revolve around concept of nature and life
- Vedic and modern science are complementary, not contradictory



The Vedas, in fact, are the oldest books in which we can study the first beginnings of our language and of everything which is embodied in all the languages under the sun.

— N.J. Lockyer, *The Dawn of Astronomy*

Vedic Concept of Environment (Paryāvaraṇa)

Chandāṁsi — Three Coverings of Our Surroundings

"Wise utilise three elements variously which are varied, visible and full of qualities. These are water, air and plants or herbs. They exist in the world from the very beginning. They are called Chandāṁsi meaning coverings available everywhere." — Atharvaveda 18.1.17



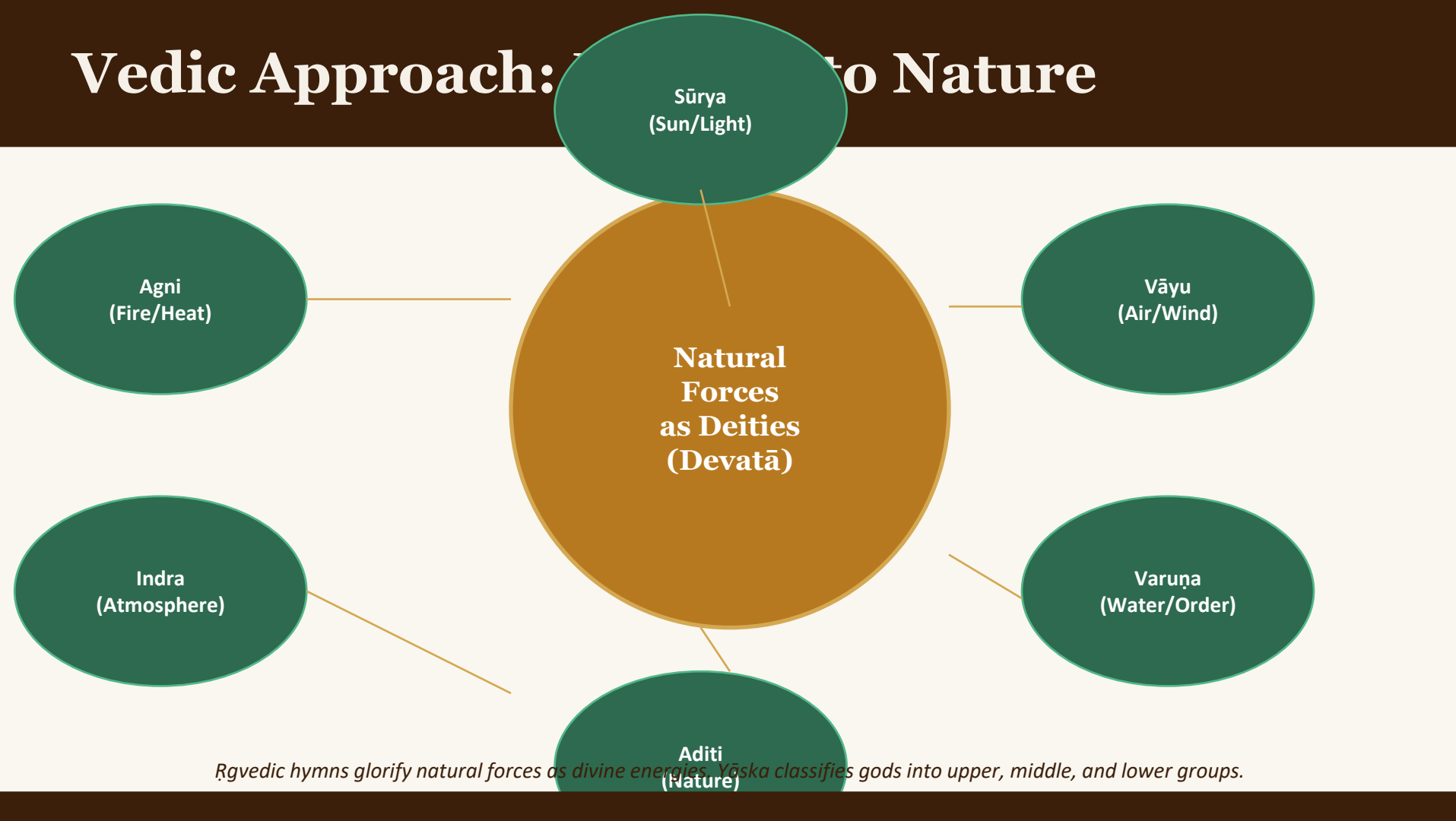
Water
(Āpas)

Air
(Vāyu)

Plants
(Oṣadhi)

The Upaniṣads describe five basic elements: Earth • Water • Light • Air • Ether

Vedic Approach: From Cosmos to Nature



Ṛta — Cosmic Order & Varuṇa

Ṛta — Universal Law

- Governs everything in the universe
- Reduces chaos to cosmos; gives order to matter
- Gives symmetry and harmony to the environment
- "No one can ignore it, even gods are abided by Ṛta"
- S.R.N. Murthy: Ṛta = a law of gravitation in simple form
- Sustains the sun in the sky and keeps the world in order

Varuṇa — Lord of Ṛta

- Sovereign god, great king, law-maker of the cosmos
- Lord of water, ocean, and universal order
- Holds heaven and earth apart
- Makes path for the sun; ordains the moon at night
- Causes rivers to flow across the land
- Represents consciousness of controlling natural forces

Division of the Universe — Tripartite Model



A vertical dashed line on the left side of the slide, with three colored rectangular blocks (orange, dark green, and olive green) positioned next to it, corresponding to the three layers of the universe model.

DYAU (Heaven/Sky)

Light Space — Dyuloka
Sūrya (Sun) rules here
Stars, moon, celestial bodies

ANTARIKṢA (Intermediate Space)

Aerial Region — between heaven and earth
Vāyu / Indra rules here
Air, storms, rain clouds

PR̥THIVĪ (Earth)

Observer Space — where we live
Agni (Fire) rules here
Herbs, plants, oceans, land

Pañcamahābhūtas — Five Basic Elements

Earth (Pṛthivī)

Mother — sustains all
life
Bhūmi-Sūkta of
Atharvaveda
Geothermal &
Geomagnetic field

Water (Āpas)

5 forms: Rain, Spring,
Canals, Lakes, Rivers
Cycle:
ocean→sky→earth

Light (Jyotis)

Sūrya — soul of all
moving
and non-moving things
Dispenser of darkness

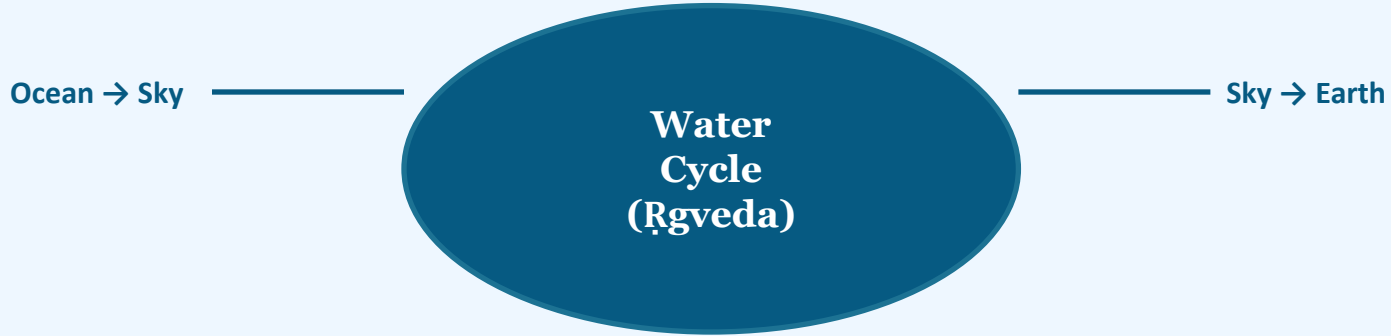
Air (Vāyu)

Soul of all deities
Life-breath inside body
Medicinal & purifying

Ether (Ākāśa)

Related to sound
Two types: inside &
outside body
Do not pollute the sky

Concept of Water (Āpas)



Rain Water

Divyāḥ

Natural Spring

Śravanti

Wells & Canals

Khanitrimāḥ

Lakes

Svayamjāḥ

Rivers

Samudrārthā

Padma Purāṇa: 'The person who pollutes waters of ponds, wells or lakes goes to hell' — Ancient water protection law

Plants, Animals & Biodiversity in Vedic Texts



Plants & Herbs (Oṣadhi)

- Oṣadhi-Sūkta of Ṛgveda addresses plants as mother
- Aranyāni — queen of the forest — praised in Ṛgveda 10.146
- "One tree is equal to ten sons" — Purāṇa
- Ṛgveda: forests should not be destroyed (8.1.13)
- Avi element = chlorophyll (cause of greenness in trees)
- "O Earth! What on you, I dig out, let that quickly grow over"
- Atharvaveda: Earth = keeper of forests, trees, and herbs



Animals & Birds

Sky Animals

Birds — part of upper environment

Forest Animals

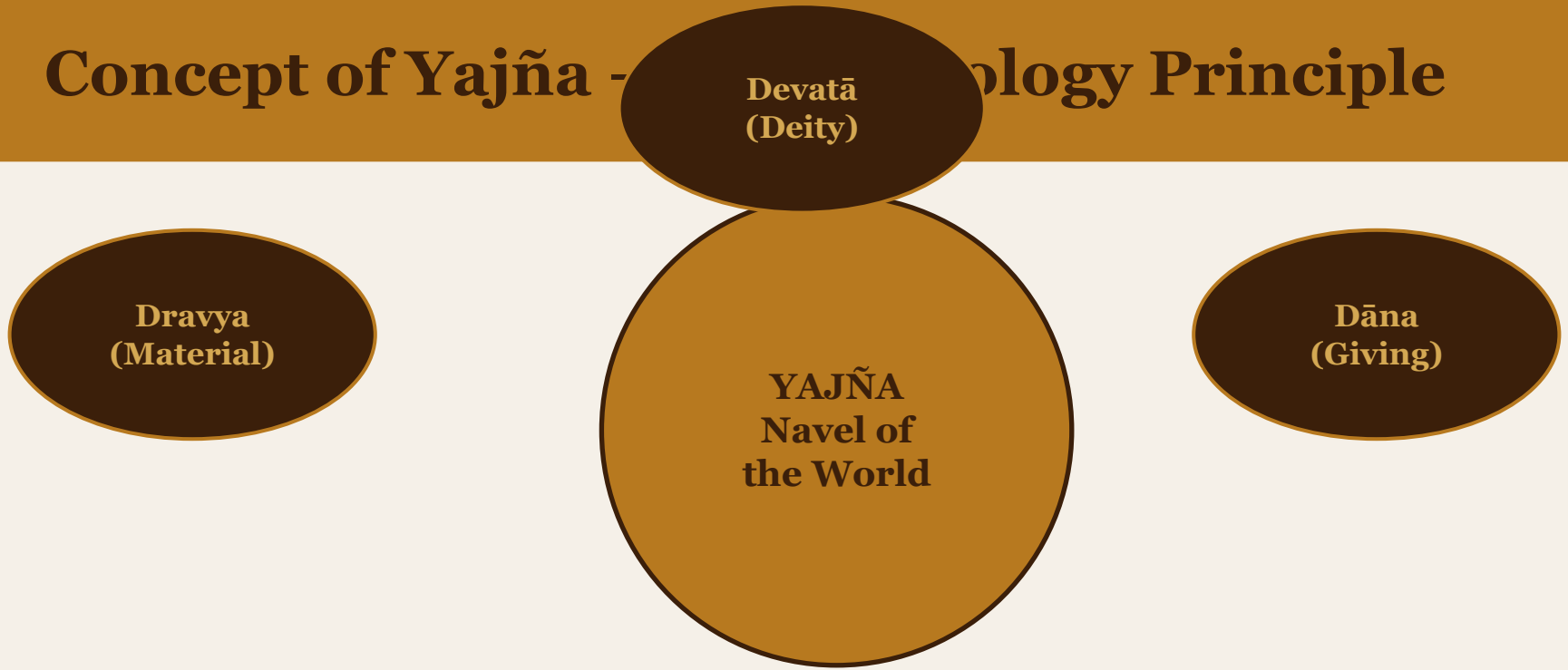
Wild animals — natural environment

Human Habitat

Domestic animals — cattle, cows

All creatures should live in peace — Yajurveda 19.20, Atharvaveda 11.2.24

Concept of Yajña - Biology Principle



Benefits of Yajña (Yajurveda):

- ♦ Minimises air pollution
- ♦ Increases crop yield
- ♦ Protects plants from disease
- ♦ Provides pure environment
- ♦ Cleanses atmosphere via medicinal smoke
- ♦ Theory of give and take in nature

Co-ordination Among Natural Powers



Śānti Mantra — Universal Peace Prayer

Peace of sky, peace of mid-region, peace of earth, peace of waters, peace of plants, peace of trees, peace of all-gods, peace of Brahman, peace of universe, peace of peace; May that peace come to me!

Sky (Dyau)

Mid-region (Antarikṣa)

Earth (Pṛthivī)

Waters (Āpas)

Plants & Trees

All Creatures

Conclusion

 Vedic vision of environment goes far beyond the physical — it is spiritual, cultural, and comprehensive.

 Ancient seers understood cosmic order (Ṛta), natural balance, and the interrelation of all elements.

 Concepts of water conservation, air purity, forest protection, and animal welfare are all rooted in Vedic texts.

 Yajña — the principle of give and take — is a major pillar of ancient Vedic environmental science.

 The Vedic message: Environment belongs to all living beings and needs protection by all, for the welfare of all.

"One should enjoy with renouncing or giving up others' part." — Īśāvāsyopaniṣad